

Model and diagnostic expressions

Level 1 — first-order linear SISO

$$G_v(s) = \frac{K_v}{\tau_v s + 1}, \quad G_r(s) = \frac{K_r}{\tau_r s + 1}$$

$$y(k) = a_1 y(k-1) + b_1 u(k-1)$$

$$G_v(s) = \frac{6.42}{1.95 s + 1}, \quad G_r(s) = \frac{0.771}{0.471 s + 1}$$

$$C(s) = \frac{\tau_v s + 1}{K_v \lambda s}, \quad k_p = \frac{\tau}{K \lambda}, \quad k_i = \frac{1}{K \lambda}$$

$$v_{\text{eq}} = K_v u$$

Level 2 — nonlinear state space with quadratic drag

$$\begin{aligned} \dot{v} &= k_T u - d_2 v |v| \\ \dot{r} &= k_\delta \delta - d_r r \end{aligned}$$

$$\begin{aligned} \dot{v} &= 3.963 u - 0.438 v |v| \\ \dot{r} &= 1.474 \delta - 1.912 r \end{aligned}$$

$$\dot{v} = -d_2 v |v|$$

$$v_{\text{eq}} = \sqrt{\frac{k_T u}{d_2}}$$

$$\frac{r}{\delta} = \frac{0.771}{0.523 s + 1} \quad \text{against Level 1's} \quad \frac{0.771}{0.471 s + 1}$$

Level 3 — residual NARX

$$\mathcal{S} = \{0, 1, 2, 5\}, \quad \mathcal{T} = \{0, 1, 2, 5, 10, 20\}, \quad \mathcal{R} = \{0, 1, 2, 5, 10\}$$

$$x(k) = \begin{bmatrix} v(k) \\ r(k) \end{bmatrix}$$

$$\varphi(k) = \left[x(k-i)_{i \in \mathcal{S}}, \quad u(k-j)_{j \in \mathcal{T}}, \quad \delta(k-l)_{l \in \mathcal{R}} \right]^\top \in \mathbb{R}^{19}$$

$$x(k+1) = x(k) + f_\theta(\varphi(k))$$

$$f_\theta(\varphi) = W_2 \tanh(W_1 \varphi + b_1) + b_2$$

$$W_1 \in \mathbb{R}^{64 \times 19}, \quad W_2 \in \mathbb{R}^{2 \times 64}, \quad 1,410 \text{ parameters}$$

The three levels side by side

$$\text{Level 1: } y(k+1) = a_1 y(k) + b_1 u(k)$$

$$\text{Level 2: } \dot{v} = k_T u - d_2 v |v|$$

$$\text{Level 3: } x(k+1) = x(k) + f_\theta(\varphi(k))$$

$$\text{Level 1: } v_{\text{eq}} = K_v u$$

$$\text{Level 2: } v_{\text{eq}} = \sqrt{k_T u / d_2}$$

$$\text{Level 3: } v_{\text{eq}} \text{ wherever } f_\theta = 0 \text{ — learned, not derived}$$

Diagnostic 1 — speed-dependent rudder gain

$$\dot{r} = k_\delta v \delta - d_r r \quad \text{against} \quad \dot{r} = k_\delta \delta - d_r r$$

$$k_\delta \propto v^2, \quad d_r \propto v$$

Diagnostic 3 — rudder-induced surge drag

$$\text{corr}(\delta^2, v - \hat{v}_{\text{Level 2}}) = -0.08$$

Diagnostic 4 — margin-based sufficiency

$$C(s) = \frac{\tau_v s + 1}{K_v \lambda s} \quad \implies \quad L(s) = \frac{1}{\lambda s}, \quad T(s) = \frac{1}{\lambda s + 1}$$

$$G_2(s, v_0) = \frac{K_2(v_0)}{\tau_2(v_0) s + 1}, \quad \tau_2 = \frac{1}{2d_2 v_0}, \quad K_2 = \frac{k_T}{2d_2 v_0}$$

$$\ell(\omega, v_0) = \left| \frac{G_2(j\omega, v_0)}{G_1(j\omega)} - 1 \right|$$

$$\|\ell T\|_\infty < 1$$

$$v_{\text{lo}} = \frac{k_T}{4 d_2 K_v} = 0.35 \text{ m/s}$$

$$\lim_{\omega \rightarrow \infty} \left| \frac{G_2}{G_1} \right| = \frac{k_T \tau_v}{K_v} = 1.20$$

$$\frac{\text{achieved bandwidth}}{\text{design bandwidth}} \text{ falls to } 0.5 \text{ at } v_{\text{hi}} = 1.28 \text{ m/s}$$

$$\text{Level 1 sufficient over } [0.35, 1.28] \text{ m/s}$$

Evaluation metric

$$R^2 = 1 - \frac{\sum_k (y_k - \hat{y}_k)^2}{\sum_k (y_k - \bar{y})^2}$$